

2.

In this question you must show all stages of your working.

Solutions relying entirely on calculator technology are not acceptable.

Determine the values of x for which

$$64 \cosh^4 x - 64 \cosh^2 x - 9 = 0$$

Give your answers in the form $q \ln 2$ where q is rational and in simplest form.

(4)

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1. (a) Using the definition of $\sinh x$ in terms of exponentials, prove that

$$4\sinh^3 x + 3\sinh x \equiv \sinh 3x \quad (2)$$

- (b) Hence solve the equation

$$\sinh 3x = 19 \sinh x$$

giving your answers as simplified natural logarithms where appropriate. (5)



1. The curve C has equation

$$y = 31 \sinh x - 2 \sinh 2x \quad x \in \mathbb{R}$$

Determine, in terms of natural logarithms, the exact x coordinates of the stationary points of C .

(7)

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Answer ALL questions. Write your answers in the spaces provided.

1. (a) Prove that

$$\tanh^{-1}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad -k < x < k$$

stating the value of the constant k .

(5)

- (b) Hence, or otherwise, solve the equation

$$2x = \tanh\left(\ln\sqrt{2-3x}\right)$$

(5)

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